Day 64 — Internal Review & System Walkthrough

Review basis

This report is a static repository review of the supplied project ZIP and the documented pipeline. No mentor/reviewer interview, live production deployment, external user study or real-candidate evaluation was supplied. Therefore, findings below are code/documentation observations and proposed actions, not stakeholder feedback.

Walkthrough

1. Resume upload / parsing: `ats_engine/api.py`, `parsers/`

2. ATS scoring: `scoring/ats_scoring_engine.py`

3. Screening: `screening_ai/`, `scoring/screening_scoring_engine.py`

4. HR interview: `interview_ai/interview.py`, `scoring/hr_interview_scoring_engine.py`

5. Technical interview: `interview_ai/technical_interview_engine.py`, `scoring/technical_skill_scoring_engine.py`

6. Machine test: `interview_ai/machine_test_engine.py`

7. Aggregation / recommendation: `scoring/cross_round_aggregation_engine.py`, `interview_ai/final_recommendation_engine.py`

Findings

| Priority | Finding | Evidence / impact | Recommended action |

| --- | --- | --- | --- |

| P0 | API stores are in-memory | `ats_engine/api.py` uses module dictionaries; restart loses state | Adopt durable DB/object storage and a persistent job queue before production |

| P0 | Applicant data requires stronger operational safeguards | resume bytes and candidate information are processed; helper audit framework is in-memory | Define consent, encryption, access, retention, deletion, incident response and audit persistence |

| P1 | ATS scorer uses falsy defaulting | `skill_match = skill_match or 0` etc.; invalid/falsy values can be silently coerced | Validate finite numeric 0–100 inputs explicitly at engine boundary |

| P1 | Several interview scoring signals are heuristic | keyword overlap, marker counts, text features | Validate against representative human-labelled data; provide evidence and reviewer override |

| P1 | Monitoring is not centrally deployed | current logger writes a local file; no exporter/dashboard/alert delivery | Integrate a shared metrics/logging platform with privacy controls |

| P2 | Recruiter-facing output consistency can vary by module | modules return different result structures | Add presentation adapter and versioned output contract |

| P2 | Full API coverage is incomplete | existing REST service focuses on ATS; interview/decision modules remain Python components | Define authenticated session APIs and durable state before integration |

Improvement action plan

• **Immediate:** validate all score inputs; ensure raw resume/transcript/answer content is never logged; test failure paths.

• **Next:** persist job state and audit metadata; add authentication, authorization and request correlation.

• **Then:** run representative validation and subgroup error analysis; document human review and appeals.

• **Before launch:** security review, privacy impact assessment, load test, backup/restore test, monitoring alert exercise and accessibility/usability review.

Acceptance criteria

• All new unit tests pass.

- Full suite passes without regressions.
- Operational metrics have agreed owners and thresholds.
- Audit trail is durable and access-controlled in deployment.
- A reviewer can understand every score's evidence and override it.